

Twenty-second Section. Abelian fields and cyclotomic fields.

§ 179. Simple Abelian fields.

We now turn to one of the most interesting applications of the theory of algebraic numbers.

The matter is the proof of the general theorem:¹

I. All Abelian number fields in the absolute domain of rationality are cyclotomic fields.

Once we have proved this theorem, the investigations of the third section on cyclotomic fields acquire an increased interest, because it is thereby shown that on the path indicated there one finds not only all cyclotomic fields, but all Abelian fields whatever.

Let

$$\Omega = R(x)$$

be an Abelian field of degree m , that is, a field consisting of all rational functions of a root x of an irreducible Abelian equation of degree m , when the field R of rational numbers is regarded as the domain of rationality. The Galois group $\mathfrak{G}$ of this equation, which is therefore an Abelian group and has the same degree m as the field Ω , is also the group of the field Ω .

If x_1 is a non-primitive number from Ω , then $\Omega_1 = R(x_1)$ is likewise an Abelian field, which we call a proper divisor of Ω , and whose degree is a proper divisor of the degree of Ω . For if $\mathfrak{A}$ is the group to which the number x_1 belongs, then $\mathfrak{G} \mid \mathfrak{A}$ (§ 4 of this volume) is the group of the field Ω_1 , and this is, together with $\mathfrak{G}$, an Abelian group. A divisor of Ω which has the same degree as Ω is identical with Ω (cf. Vol. I, §§ 144, 149, 156).

If there are now two non-primitive numbers x_1, x_2 in Ω of such a kind that

$$\Omega = R(x_1, x_2)$$

can be put, then the fields $\Omega_1 = R(x_1)$, $\Omega_2 = R(x_2)$ are proper divisors of Ω , and Ω is called *composed* from the two fields Ω_1, Ω_2 .

If the field Ω permits no such representation, it is called *simple*.

Thus, if it is assumed as proved that Ω_1, Ω_2 are cyclotomic fields, that is, that all their numbers are representable rationally by roots of unity, then the same follows for Ω .

If one of the fields Ω_1, Ω_2 is decomposable, then we can repeat the decomposition, and, since the degrees are decreasing positive integers, we must finally arrive at simple fields.

Theorem I therefore needs to be proved only for simple Abelian fields.

It is, however, to be recalled that the decomposability of a field Ω by no means already follows from the existence of a divisor, and that in a decomposable field the decomposition into simple fields can occur in quite different ways, so that for fields the laws of divisibility do not hold as they do in the domain of numbers.

A criterion for the simplicity of a field can be derived from its group.

If the group $\mathfrak{G}$ has two proper divisors $\mathfrak{A}$ and $\mathfrak{B}$ of such a kind that every element can be composed in one and only one way from an element of $\mathfrak{A}$ and an element of $\mathfrak{B}$, then we call the group $\mathfrak{G}$ decomposable into the two components $\mathfrak{A}, \mathfrak{B}$; thus $\mathfrak{G}$ is decomposable when, in the product formed according to the composition of parts (§ 4),

$$\mathfrak{G} = \mathfrak{A}\mathfrak{B}, \tag{1}$$

¹Kronecker first stated this theorem, but did not publish a complete proof of it. The first proof was made known by the author of the present work in vol. 8 of *Acta Mathematica* (1886). Very recently a second proof by Hilbert has been published, which is based on the theorems developed in the nineteenth section (*Nachrichten d. Gesellschaft d. Wissenschaften zu Göttingen*, 1896, issue 1).

each element of $\mathfrak{G}$ appears once and only once. The degree of $\mathfrak{G}$ is then equal to the product of the degrees of $\mathfrak{A}$ and $\mathfrak{B}$.

If there are no such divisors, the group $\mathfrak{G}$ is called indecomposable.

Then we can prove the theorem:

1. If the group of an Abelian field is decomposable, then the field too is composed.

To prove this, assume that the group $\mathfrak{G}$ of the field Ω is decomposable according to formula (1), and let m, a, b be the degrees of $\mathfrak{G}, \mathfrak{A}, \mathfrak{B}$. We take a number ξ of the field Ω belonging to the group $\mathfrak{B}$, which is therefore transformed by the substitutions of $\mathfrak{G}$ into a different conjugate values, and likewise a b -valued number η belonging to $\mathfrak{A}$. We can then form a number

$$x = \alpha\xi + \beta\eta$$

with rational numerical coefficients α, β , which has m different conjugate values, and then $\Omega = R(x)$. Thus Ω is composed from $R(\xi)$ and $R(\eta)$ (Vol. I, § 143).

If we now recall the representation of an Abelian group by a basis derived in §§ 9, 10 of this volume, then repeated application of what has just been proved gives:

2. Every Abelian field Ω can be composed from such fields whose degree is a power of a prime number and whose group is cyclic. The degrees of these composing fields are the invariants of the group of Ω .

These theorems are supplemented by the following:

3. An Abelian field of prime-power degree with cyclic group is simple.

If the group $\mathfrak{G}$ is cyclic, then its elements can be represented by the powers of a basis, G^γ , when γ runs through a complete system of residues modulo m . We obtain all divisors $\mathfrak{A}$ of $\mathfrak{G}$ represented by

$$\mathfrak{A} = G^{b\gamma},$$

when b is a divisor of m and γ runs through a complete system of residues modulo $a = m : b$.

If

$$\mathfrak{A}' = G^{b'\gamma}$$

is a second divisor of $\mathfrak{G}$ of degree a' and if

$$b' \leq b, \quad a' \geq a,$$

then, when we now assume that m and hence also a, b, a', b' are powers of a prime number q , b' is a divisor of b , and consequently $\mathfrak{A}$ is a divisor of $\mathfrak{A}'$.

If therefore ξ is a number of the field Ω which belongs to the group $\mathfrak{A}$, then ξ is a primitive number of the field $R(\xi)$ of degree b , and in this field every number ξ' is also contained which admits the substitutions of the group $\mathfrak{A}'$. Thus if $\mathfrak{A}, \mathfrak{B}$ and $\mathfrak{A}', \mathfrak{B}'$ are proper divisors of $\mathfrak{G}$, then $R(\xi)$ is different from Ω , and $R(\xi)$ is not enlarged by the adjunction of ξ' . Hence Ω also cannot be composed from $R(\xi)$ and $R(\xi')$, whereby 3 is proved.

§ 180. The resolvents.

According to what has now been proved, in the further considerations, whose aim is the proof of Theorem I, we may restrict ourselves to such Abelian fields whose degree is a prime power and whose group is cyclic. For this reason it also sufficed, for the intended application, that in the preceding section we restricted ourselves to the consideration of the cyclotomic fields

Ω_m in which m was a prime power. We retain as much as possible of the notation used there and put

$$m = q^\kappa, \quad (1)$$

where q is a prime number and κ a positive exponent; and in the case where $q = 2$ we assume $\kappa > 1$. By n we denote any number not divisible by q . Let r be a primitive m -th root of unity, and Ω_m the field of rational functions of r .

Let now $\Omega = R(x)$ be a simple Abelian field of degree m , so that x is a root of a rational irreducible cyclic equation of degree m . We denote the roots of this equation, in a suitable order, by

$$x_0, x_1, x_2, \dots, x_{m-1}, \quad (2)$$

and assume $x_m = x_0$, $x_{m+1} = x_1$, and so on. Then

$$x_1 = \Phi(x_0), \quad x_2 = \Phi(x_1), \dots, \quad x_0 = \Phi(x_{m-1}),$$

where Φ denotes an integral function with rational coefficients, and the group of the field Ω consists of the powers of the substitution (x_0, x_1) , or of the cyclic permutations

of the roots (2). Every cyclic function of these quantities is a rational number. What we have to prove is that the x can be expressed rationally by roots of unity. If we adjoin to the field Ω the m -th root of unity r , there arises a field $R(x, r)$ which contains the two fields $\Omega = R(x)$ and $R(r) = \Omega_m$ as divisors.

If a number $\omega = \Phi(x, r)$ of the field $R(x, r)$ admits all substitutions (x_0, x_1) , (x_0, x_2) , $\dots$, (x_0, x_{m-1}) , then it is contained in the field Ω_m ; for then

$$m\omega = \Phi(x_0, r) + \Phi(x_1, r) + \dots + \Phi(x_{m-1}, r),$$

and the coefficients of the powers of r in this expression are not only cyclic but even symmetric functions of the roots (2), hence rational numbers. This remains true even in the case where the equation for x is reduced by adjunction of r .

We apply this to the resolvents

$$(r, x_0) = x_0 + rx_1 + r^2x_2 + \dots + r^{m-1}x_{m-1}, \quad (3)$$

and, more generally, when s denotes an arbitrary integral exponent,

$$(r^s, x_0) = x_0 + r^sx_1 + r^{2s}x_2 + \dots + r^{(m-1)s}x_{m-1}. \quad (4)$$

By these resolvents x_0 itself can again be rationally expressed, namely

$$mx_0 = \sum_{s=0}^{m-1} (r^s, x_0), \quad (5)$$

and if therefore we can show that all these resolvents (r^s, x_0) are cyclotomic numbers, then we have reached our goal. If we put

$$(r^s, x_\lambda) = x_\lambda + r^sx_{\lambda+1} + r^{2s}x_{\lambda+2} + \dots + r^{(m-1)s}x_{\lambda+m-1}, \quad (6)$$

then (r^s, x_λ) arises from (r^s, x_0) by the substitution (x_0, x_λ) , and the fundamental relation holds:

$$r^{\lambda s} (r^s, x_\lambda) = (r^s, x_0). \quad (7)$$

All these resolvents are numbers of the field $R(x, r)$; we consider preferably the following two combinations:

$$(r^s, x_0)^m = \omega_s, \quad (8)$$

$$(r^s, x_0)^{-1}(r, x_0)^s = \alpha. \quad (9)$$

These two numbers admit, as follows immediately from (7), the substitutions (x_0, x_λ) and are therefore numbers of the field Ω_m .

If n is not divisible by q , then, because of the irreducibility of the cyclotomic equation, none of the resolvents (r^n, x_0) can vanish unless they all vanish; for the equation

$$(r, x_0)^m = 0$$

would be an equation for r with rational coefficients, admitting all substitutions (r, r^n) .

Likewise, none of the resolvents (r^s, x_0) can vanish unless the whole family vanishes in which s has a fixed greatest common divisor with m .

If all (r, x_0) vanish, formula (9) is not applicable. Then, however, we can directly consider the resolvents (r^q, x_0) , which belong to a field of degree mq^{-1} arising from

$$y_0 = x_0 + x_q + x_{2q} + \cdots + x_{m-q},$$

and if these quantities too vanish, we pass to the resolvents (r^{q^2}, x_0) , and so on.

If, however, (r, x_0) does not vanish, then by formula (9) all (r^s, x_0) can be represented rationally by (r, x_0) and by cyclotomic numbers.

Consequently it is sufficient for our purpose if we can prove that the non-vanishing resolvents (r, x_0) are cyclotomic numbers.

Let n run through a complete system of incongruent numbers not divisible by q . Then

$$\sum n \equiv 0 \pmod{m},$$

for the numbers n split into pairs which complement one another to m . From (7) it follows that the product

$$\prod_n (r^n, x_0) = a \quad (10)$$

is a number in Ω_m . But since this number moreover is not changed when any of the substitutions (r, r^n) is performed in it, a is a rational number.

§ 181. Preparation for the proof.

After the developments of the preceding paragraph, for the proof of the great Theorem I it remains only to show that the number contained in Ω_m ,

$$\omega = (r, x_0)^m,$$

is the m -th power of a cyclotomic number.

1. The essential property of the numbers ω_n conjugate to ω , on which the proof rests, is that

$$\omega_n^{-1} \omega_1^n = \alpha^m \quad (1)$$

is the m -th power of a number in Ω_m .

Here n may be any number not divisible by q . If we form, of the right- and left-hand sides of (1), the norms in the field Ω_m , then, denoting by a the norm of the number α , which is a rational number, and observing that every ω_n has the same norm as ω , we obtain

$$N(\omega)^{n-1} = a^m.$$

If now m is odd, then we can put $n = 2$ in this, and obtain as a consequence of 1:

2. The norm of ω is the m -th power of a rational number,

$$N(\omega) = a^m. \quad (2)$$

If m is a power of 2, this inference is no longer permissible. It then follows from 1 only that $N(\omega)^2$ is the m -th power of a rational number. Nevertheless, as is seen from (10) of the preceding paragraph, the numbers ω must, also in this case, still satisfy condition 2; but this condition is then no longer contained automatically in 1, and is added as a new property.

What we have to prove is that from 1 and 2 it follows that ω itself is the m -th power of a cyclotomic number, even if in a higher field.

For this it is necessary to investigate the decomposition of the numbers ω into their prime factors in the field Ω_m .

The prime number q is, as was proved in § 169, associated with the $\varphi(m)$ -th power of a prime number $\sigma = 1 - r$ existing in Ω_m .

If therefore σ^k is the power of σ contained in ω , we put

$$\omega = \sigma^k \omega',$$

where ω' is relatively prime to the prime number q . Formation of the norm gives, by 2,

$$a^m = q^k b,$$

where b is a rational number which, in lowest terms, contains the prime number q neither in the numerator nor in the denominator. From this it follows that k must be divisible by m , and hence the first result:

3. The exponent of the prime number σ contained in ω is always divisible by m .

Let now p be a prime number different from q , and let $\mathfrak{p}_\xi$ be a prime factor of p in the field Ω_m . Let ξ run through the sequence of numbers $\xi_1, \xi_2, \dots, \xi_e$ determined more precisely in § 172 in the different cases, so that

$$p = \mathfrak{p}_{\xi_1} \mathfrak{p}_{\xi_2} \cdots \mathfrak{p}_{\xi_e}, \quad (3)$$

and assume that $\mathfrak{p}_\xi^{k_\xi}$ is the power of $\mathfrak{p}_\xi$ contained in ω . The power of $\mathfrak{p}_{n\xi}$ contained in ω_n is then $\mathfrak{p}_{n\xi}^{k_\xi}$; and if we determine n' from the congruence

$$nn' \equiv 1 \pmod{m}, \quad (4)$$

then $\mathfrak{p}_\xi^{k_{n'\xi}}$ is the power of $\mathfrak{p}_\xi$ contained in ω_n . Accordingly, in

$$\omega_n^{-1} \omega_1^n \quad (5)$$

the power $\mathfrak{p}_\xi^{nk_\xi - k_{n'\xi}}$ is contained, and by 1 the exponent of this power must be divisible by m .

Thus for every n not divisible by q we obtain the congruence

$$nk_\xi \equiv k_{n'\xi} \pmod{m}, \quad (6)$$

in which we can now also replace ξ by $n\xi$, whereby

$$nk_{n\xi} \equiv k_\xi \pmod{m} \quad (7)$$

is obtained. If in this we take $\xi = 1$ and put ξ, ξ' for n, n' and k for k_1 , then from (7) follows

$$k_\xi \equiv \xi' k \pmod{m}, \quad (8)$$

where k now has the same value for all ξ , and ξ, ξ' are connected by the congruence

$$\xi\xi' \equiv 1 \pmod{m}.$$

If we take $n = p$, then $\mathfrak{p}_{p\xi} = \mathfrak{p}_\xi$ (§ 172), and so also $k_{p\xi} = k_\xi$; and from (7) and (8) follows

$$k(p-1) \equiv 0 \pmod{m}. \quad (9)$$

Important consequences now follow from this. First, if $p-1$ is not divisible by q , then it follows that k must be divisible by m , and we therefore have the theorem:

4. The prime factors of a prime number p which is not congruent to 1 modulo q are contained in ω only in such powers whose exponents are divisible by m .

If, in the case where m is a power of 2, $p-1$ is divisible by 2 but not by 4, that is, if $p \equiv -1 \pmod{4}$, then it follows from (9) only that k is divisible by $\frac{1}{2}m$, not that k is divisible by m . The exponent of the power of $\mathfrak{p}_\xi$ contained in ω is, apart from multiples of m , equal to k ; and if k is divisible by m , everything is as in case 4. But if k is divisible only by $\frac{1}{2}m$, not by m , so that

$$k \equiv \frac{1}{2}m \pmod{m},$$

then $k - \frac{1}{2}m$ is divisible by m , and with reference to the decomposition (3) we can formulate the theorem as follows:

5. If m is a power of 2 and p is a prime number congruent to -1 modulo 4, then the exponent h can be so determined that all prime factors of p in

$$\omega\sqrt{p}^{-hm}$$

are contained only in such powers whose exponent is divisible by m .

Finally let $p-1$ be divisible by q , or, if m is even, by 4. Let m_1 be the greatest common divisor of m and $p-1$, and let

$$m = m_1 m_2.$$

Then by (9) k must be divisible by m_2 ; and if we put

$$k = h m_2,$$

then by (8) the product contained in ω is

$$\left(\prod_{\xi} \mathfrak{p}_{\xi}^{\xi'} \right)^{h m_2}. \quad (10)$$

In addition, the prime factors $\mathfrak{p}_\xi$ occur in ω only in such powers whose exponents are divisible by m .

In this case ξ , by § 172, III, runs through a complete system of residues not divisible by q modulo m_1 , and in (10) we may understand by ξ' the least positive solution of the congruence

$$\xi\xi' \equiv 1 \pmod{m_1},$$

because, if we change the exponents ξ' in (10) modulo m_1 , only m -th powers of the prime factors $\mathfrak{p}$ are added.

Then we can apply Kummer's theorem [§ 174, (16)] to the product (10), replacing m there by m_1 , and obtain, if we understand by $r_1 = r^{m_2}$ an m_1 -th root of unity and by the η certain periods of the p -th roots of unity,

$$\prod_{\xi} \mathfrak{p}_{\xi}^{\xi'} = (r_1, \eta)^{m_1},$$

and consequently

$$\left(\prod_{\xi} \mathfrak{p}_{\xi}^{\xi'} \right)^{hm_2} = (r_1, \eta)^{hm}. \quad (11)$$

Consequently we have the theorem:

6. If p is a prime number and $p - 1$ is divisible by q , or, when $q = 2$, by 4, then the positive exponent h can be so determined that the number contained in Ω_m

$$\omega(r_1, \eta)^{-hm}$$

contains the prime factors of p only in such powers whose exponent is divisible by m .

It is now also no longer necessary to distinguish Theorem 5 from Theorem 6; for for $m_1 = 2$, 6 passes into 5, because then $r_1 = -1$, and by Vol. I, § 171,

$$(r_1, \eta)^m = (-1, \eta)^m = \sqrt{p^m}$$

holds.²

If we collect the result of Theorems 3 to 6 in one formula, we obtain, when ε denotes a unit functional, φ any functional of the field Ω_m , and p runs through a sequence of prime numbers congruent to 1 modulo q , among which the same prime number p may occur several times,

$$\omega = (r, x_0)^m = \varepsilon \varphi^m \prod_p (r^{m_2}, \eta)^m. \quad (12)$$

The resolvents of the cyclotomy occurring in these formulae, (r^{m_2}, η) , are cyclotomic numbers in a higher field. But their m -th powers are numbers contained in the field Ω_m which are transformed by the substitution (r, r^n) into

$$(r^{nm_2}, \eta)^m.$$

Likewise the combinations

$$(r^{nm_2}, \eta)^{-1} (r^{m_2}, \eta)^n \quad (13)$$

are contained in the field Ω_m [§ 174, (9), (11)]. These resolvents are special cases of the general resolvent (r, x_0) . From (12), if we denote by φ_n, ε_n the functionals conjugate to φ and ε , we obtain

$$\omega_n = \varepsilon_n \varphi_n^m \prod_p (r^{m_2 n}, \eta)^m. \quad (14)$$

This formula teaches us the most important properties of the functionals φ_n and ε_n , first:

7. The functionals φ_n^m are associated with numbers of the field Ω_m , and therefore belong to the principal class (§ 152).

²In the author's paper in vol. 8 of *Acta Mathematica* it was omitted to emphasize separately the case of Theorem 5.

By 1, $\omega_n^{-1}\omega_1^n = \alpha^m$ is the m -th power of a number of Ω_m . Since the combinations (13) are also numbers in Ω_m , we can put

$$\prod_p (r^{nm_2}, \eta)^{-1} (r^{m_2}, \eta)^n = \frac{\alpha}{\beta},$$

and obtain from (14)

$$\varepsilon_n^{-1} \varepsilon_1^n (\varphi_n^{-1} \varphi_1^n)^m = \beta^m,$$

where β is a number in Ω_m .

It follows from this that the product

$$\beta \varphi_n \varphi_1^{-n} = \varepsilon$$

is a unit functional of the field Ω_m , and from this arise, for the functionals φ_n and the unit functionals ε_n , the following two theorems:

8. The conjugate functionals φ_n have the property that all products

$$\varphi_n \varphi_1^{-n}$$

belong to the principal class.

9. The unit functionals ε_n have the property that the products

$$\varepsilon_n \varepsilon_1^{-n} = \varepsilon^m$$

are m -th powers of unit functionals.

For the proof of the principal theorem I, everything now depends on the proof of the two propositions which are to be inferred from the properties of the functionals φ and ε :

A. The functionals φ_n belong to the principal class.

If we may presuppose this lemma, then φ_n is associated with a number α_n , and we can write formula (14), with a somewhat changed meaning of ε_n , in the form

$$\omega_n = \varepsilon_n \alpha_n^m \prod_p (r^{m_2 n}, \eta)^m. \quad (15)$$

In this formula, however, ε_n is a numerical unit of the field Ω_m satisfying theorem 9.

If, therefore, from theorem 9 we can still prove the second lemma:

B. A numerical unit ε_n of the field Ω_m , which satisfies theorem 9, is the m -th power of a unit in Ω_m , multiplied by a power of r ;

then in (15) we can extract the m -th root, and obtain, when by ρ a root of unity of possibly higher degree than m and by α a number of the field Ω_m is denoted,

$$(r, x_0) = \rho \alpha \prod_p (r^{m_2}, \eta), \quad (16)$$

where now on the right there stand nothing but cyclotomic numbers, and by this Theorem I is therefore proved.